

SQL investigation report

Northstar Solutions

The security operations team has received reports of unusual authentication behaviour. No confirmed breach has been declared at this time. My task has been to do an initial review of the authentication records and determine whether any activity requires escalation.

During my investigation I reviewed two datasets in particular. The employee records and login events. The authentication dataset covered the period from 17 August to 23 August 2026 and contained 1,142 authentication events across 60 employee accounts. The investigation was conducted with DB browser for SQL lite.

Through various functions, Analysis focused on identifying authentication anomalies based on failure volume, timing, geographic origin, device type and source IP address. Accounts exhibiting anomalous behaviour were subjected to further analysis to determine whether escalation was required..

After checking for failed login attempts, I searched the database for other suspicious login activity including access from an employee currently listed as terminated from the company. Below are my findings for the two individual incidents.

Finding 1 - Suspected account compromise

My first query in the database was regarding high amounts of failed login attempts.

```
SELECT e.first_name, e.last_name, e.employment_status, e.username, COUNT(l.success) AS failed_logins
FROM employees AS e
LEFT JOIN login_events AS l
ON e.username = l.username
AND (l.success=0)
GROUP BY e.first_name, e.last_name, e.employment_status, e.username
ORDER BY failed_logins DESC;
```

The query established failed authentication activity across all 60 employee account. Results were then ordered by number of failures descending, allowing me to see the accounts with the highest failures first. The account user mmoore was then identified as having 30 failed logins and warranted further investigation.

Upon further investigation, an IP address of **91.214.124.77** with **Russia** as the country of origin was the leader of these failed attempts. This immediately raised suspicions and therefore lead to further investigation.

From the below query I was able to ascertain the following:

Time	Activity	Assessment
01:12	Authentication attempts begin from 91.214.124.77	Anomalous source
01:12-02:10	28 failed attempts from IP address	Potential brute force attack
02:10	Successful authentication	Potential compromise
02:19	Priveleged-access event	Increased potential impact

```
SELECT timestamp, username, success, event_type
FROM login_events
WHERE ip_address = '91.214.124.77'
ORDER BY timestamp ASC;
```

I verified further details for the account, ascertaining that the user is based in London in the HR department, regularly authenticating with an iPhone. This further solidified the suspiciousness of the IP address, given it was located in Russia and gaining access with a Linux device.

Nothing further occurred with this IP address following the privileged access event, and further, this IP address did not attempt to access any other accounts.

My recommendations following this aspect of my investigation would include:

1. The user is made aware of the attempts to access their account and they are required to change their password immediately. Stronger password choice should be implemented including a potential multi-factor authentication, i.e a code send to a phone or biometrics.
2. User access should be checked on the account to ensure that they mohave least privilege required for the role.

3. Further investigation is conducted into the account. The databases do not show what was accessed during the privileged access event, we would need to determine if any files or sensitive data was accessed.

Finding 2 Potential account termination issue

Following the above incident, I continued to consult the databases to check for any other anomalies. During this, I identified that a user who was listed as terminated from the company was both still on the system and also had accessed the system successfully.

```
SELECT l.*  
FROM employees AS e  
LEFT JOIN login_events AS l  
ON e.username = l.username  
WHERE employment_status = 'Terminated'  
ORDER BY timestamp ASC;
```

The above query combined my databases, showing me the employee's currently listed as terminated and if they had accessed the system. This returned with a username rshah, having accessed the system on 22 August 2026 at 10:14 using a Windows laptop.

Unfortunately the databases do not show when the employee was terminated and this would therefore need to be clarified. If the user was terminated after the above date then there is no issue presented, however if they were terminated before 22 August 2026, it would need to be further investigated in order to determine what files or data was accessed.